

DATA MANAGEMENT PLAN

PI Gu

The Gu Laboratory is committed to the open sharing and prompt dissemination of all data and resources generated during this project. We will adhere to policies for data management and data sharing set by the NSF.

Proteomic Data

Description

This project will generate PL-LFQMS profiling data using 13 newly identified PNET proteins as well as KAKU4 and Nup188 as bait. In addition, we will perform substrate-trap PL-LFQMS profiling using two INM PNET proteins to identify potential candidates of PIIIs.

Archiving and Dissemination

We will make the raw proteomic data available to the public as soon as it has passed quality control criteria in our hands. These data will be deposited to the ProteomeXchange Consortium, which has been set up to provide a coordinated submission of MS proteomics data to the main existing proteomics repositories. All submissions will be set as publicly accessible and include metadata such as transgenic background, bait protein, tag information, controls, etc. Prior to the submission to ProteomeXchange Consortium, the MS data will be maintained both on local hard drives and in the CyVerse Data Store. CyVerse is an NSF-funded cyberinfrastructure initiative dedicated to making large scale computational resource available to a broad range of scientists.

Normalized MS data, differential enrichment analyses, and associated statistics will be published in peer-reviewed journals and provided as supplementary tables. All analyses and results in papers will also be made freely available upon request following their publication.

Mutants, Transgenic Lines, and Plasmid Constructs

Description

This project will generate biological materials of use to the research community, including Arabidopsis transgenic lines expressing fluorescent protein- and proximity biotinylation enzyme-tagged NE proteins, Arabidopsis mutants in which NE-associated genes have been disrupted through CRISPR-based genome editing, as well as plasmid constructs for expression of NE-associated Arabidopsis genes.

Archiving and Dissemination

All plasmids constructed will be stored as glycerol stocks in *E. coli* or *A. tumefaciens* in -80°C freezer. All transgenic and mutant seeds generated will be maintained in the lab in desiccated environment at room temperature. All above seeds and DNA constructs will also be deposited into publicly accessible repositories, such as Addgene and the Arabidopsis Biological Resource Center (ABRC) at Ohio State University, which will make them available to the research community. They will also be freely available upon request to PI Gu.

Publications

Description

Data obtained in the project will be published promptly. We anticipate that publication of results obtained under this project will start at Year 4 and be completed within a year following the end of the supported period. All data will be made freely available upon request following their publication. There are no

anticipated ethical or privacy issues, permission restrictions, or intellectual privacy issues associated with the data following their publication. The PI and the two main participants of this project are also expected to present their findings at two major scientific conferences and other local scientific events.

Archiving and Dissemination

In accordance with the University of California Open Access Policy, publications will be uploaded to eScholarship, UC's open access repository, and publishing platform, to be made available to the public at no charge.

Teaching Curricula

Description

Lectures on the NE evolution, structure, composition, and function will be developed. In addition, URAP and SPUR student training plans will be generated.

Archiving and Dissemination

Lectures, student presentations, data collection, training resources will be maintained online through the UC Berkeley bCourses course management system and backed up online through the unlimited cloud-based storage space offered by Google.